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RESEARCH ARTICLE

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Enhancement of photovoltaic power farms using a new power prediction approach

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Summary

Solar energy power prediction is a vital process for designing and operating solar power plants to evaluate the system performance and reduce the cost of grid management as it decreases the grid imbalance for a more stable grid. This study introduces a new power prediction approach to enhance the power prediction quality by combining different solar models. The study investigates the performance of 4.5 MW_p solar field located at the California Polytechnic State University at San Luis Obispo, California, United States to validate this new approach. The system is real-time monitored for 1 year to study its performance. A representative identical model for the actual farm is then designed and built on a solar performance modeling software to validate the software. The actual energy generation of the field is compared with the predicted energy. The model is optimized by selecting different transposition and decomposition models to achieve the most accurate energy prediction. After that, the model is used in a computational framework to predict the energy injected into the grid for photovoltaic solar farms in Egypt based on the new approach methodology. Egypt is considered a solar belt country. Following the validation, a pilot model is used to predict energy generation for the selected locations in Egypt. Egypt's case study is then used as a pilot project to optimize some design parameters to study its effect and the suitability of the power generation in Egypt's weather. The results show that the relative error using the new approach for the whole year when using the proposed approach of combining the solar models is 2.9%. The performance ratio of the actual farm is 79% for the whole year, and the total energy produced is 10.92 GWh/year. The power prediction for the Egyptian sites concludes that 14 GWh/year energy potential can be achieved from the Red Sea location from the suggested model and the most suitable modules are polycrystalline. Furthermore, predicting a 13% increase in power generation due to using tracking technology compared with the usage of polycrystalline with fixed mounts.

KEYWORDS

modeling software, performance analysis, photovoltaic power plants, PlantPredict, power prediction, solar energy

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1 | INTRODUCTION

Solar energy is one of the most promising energy sources available, but a challenge is that it fluctuates due to cloud coverage and can cause an imbalance on the electrical grid. The grid requires balanced energy sources to operate in a stable fashion and respond to consumer demand. Thus, power prediction is the world concern nowadays, especially in countries with strategic plans to implement huge solar energy projects.

Egypt is a key country in the continent because it has huge solar energy potential. It is considered a Sun Belt country because it has direct solar irradiance between 2000 and 3000 kWh/m^2/year with an average of 9 to 11 hours per day of sunshine according to Ref. 1. This solar energy potential is critical for a gradual move toward independence from fossil fuels. Egypt's expanding economy, rapid population growth, and bull market require an increase in energy generation. For those reasons, the Egyptian government has set several goals to meet the gradual increase in energy production and to use that energy for more economic development.[2] The government aims to achieve short-term objectives to increase renewable energy electricity production to at least 50% of the total national energy production. "Egypt has developed an overarching regulatory framework for developing renewable energy capacity with the aim of securing a 20% of its energy generation from renewable sources by 2022."[3] It is critical for Egypt today to be able to predict its potential for solar energy before installing a solar field. This prediction can be done using available simulation software and identifying the best location with accurate energy prediction. The power prediction model would help improve the accuracy of power generation projection that would help in the decisions of activating and deactivating fossil fuel facilities if needed for grid management.[4] It would also be an important tool to predict the power generated from a solar site before its construction. The variability and uncertainty of solar energy cause strain on the grid. This issue led to much research in the grid management sector. Lolla et al[5,6] conducted a comprehensive review of recent trends in power management and optimum generation scheduling to achieve optimum operation of distributed energy resources in microgrids. In the study by Kumar et al,[7] they have proposed a stochastic energy management system framework using an intelligent demand-side management approach for optimum energy scheduling.

The literature shows that different PV simulation software packages (such as PVsyst, PolySun, SAM and PVSST) have been used for this type of investigation. The accuracy of six different photovoltaic software packages for planning and performance estimation of the installations is studied by Sobri et al.[8] They made comparisons between the estimations and the actual electrical energy generated by the solar fields. The software programs investigated are TRYNSYS, Archelios, Polysun, PVSyst, PV*SOL, and PVGIS. By comparing the actual and estimated power generation, they have found that the software overestimates the solar irradiation received by the PV panels, even though they underestimate the electrical energy generated. The errors have been compared by four different statistical parameters, which are RMSE, MAD, MAPE, and EF. The electricity generated calculation error is mainly due to the model of the PV panels. Wijeratne et al[9] have studied 23 PV design and management software packages and four mobile applications. Fifteen subfactors for the selection of PV system installation are investigated, consisting of geophysical, technical, economic, and environmental factors. After examining the software and applications, they have identified 14 issues related to these tools. Chikh et al[10] have studied a simulation tool for PV systems, which is PVSST1.0, and compared it with PVsyst. The software focuses on the sizing part. They concluded that the results are acceptable, although the software needs some improvements.

For the PV power prediction[11] used PVsyst software for the stand-alone photovoltaic system by predicting 1-year energy output. It is concluded that PVsyst software enables solar PV system designers to design and predict the energy production based on site location and system sizing. Furthermore, Kumar et al[12] have studied the feasibility of installing a photovoltaic system in an educational institution using a PVsyst simulated model. The weather data are collected from Meteonom7.1 data sets and used in the PVsyst model. By analyzing the performance of this system, they concluded the feasibility of the installation and calculated the annual PR of 80% and computation of the loss diagram. Monsalve et al[13] investigated the installation of solar PV panels on buildings at the KTH campus in Stockholm, Sweden. They chose two buildings to be simulated with PVsyst. The results show that the energy production predicted by arcGIS tools is higher than the energy predicted from PVSyst. They also found that the PV installation would not cover the energy consumption of the campus. A similar study by Malvoni et al[14] investigated the performance of a 960 kW PV system in the Mediterranean climate of Italy. Their methodology is to monitor data over a 43-month period. They carried a comparison in terms of degradation rate with other installed sites in different climates to determine the crucial parameters. They used two PV simulation software packages, which are PVsyst and SAM. They concluded that an increase of 10 W/m^2 plane of array irradiance makes an increase of 0.26 kWh/day in the

Mediterranean climate and 1°C rise in temperature led to a loss of energy up to 7%. The simulation tools SAM overestimated the performance ratio as well as the capture losses and underestimated the AC energy, while PVsyst underestimated the PR and overestimated the system and capture losses. Ref. 15 developed an upscaling method for estimating the photovoltaics in a small area of Italy based on the clustering of PV fleet and neural network models. For the use of simulation tools to show the energy potential[16] have practically studied photovoltaics in buildings in rural England region and studied a simulated model in Malaysia. The actual data in England are compared with the simulated model in Malaysia using "PVSYST2.0" prediction, which is enhanced with the "SUNREL1.0β" thermal computer model. They concluded that cost feasibility was not effective in the United Kingdom at that time, while the simulation in Malaysia shows good potential. Moslehi et al[17] have evaluated different power prediction models for PV systems. They investigated the effect of the module temperature, climatic variables, and independent variables, such as wind speed and incident angle modifier, on the models' accuracy. They identified different scenarios depending on the type of climatic and PV data used. When only climatic data were used, the RMSE of the hourly prediction was between 3.2% and 8.6%. Roumpakias and Stamatelos[18] proposed a performance analysis procedure during 3 years period by a comparative analysis of grid-connected PV system. Their aim is to evaluate three existing models and calculate the PR. It is concluded that there is a small drop in PR from 2013 to 2015; the Evans model overestimates the production, while the improved bilinear model underestimates the production. While some studies evaluated the performance of the PV systems worldwide, these studies are summarized in Table 1.

According to this literature review, this study proposes a new approach to enhance the solar prediction quality for PV projects selection and operation phases. This approach depends on combing different transposition and decomposition models to increase the prediction quality to cover the maximum number of solar parameters. To validate this new approach, an actual field is needed, therefore "Gold Tree Solar Farm" is selected, monitored, and its performance is analyzed for the first time since the installation. Moreover, there is a gap in the current research to apply an enhanced solar power prediction and identify the best design parameters for future projects in Egypt. Thus, after validating the approach, the work implements this approach to different locations for current and future projects in Egypt, in addition to studying the design parameters to fill this gap. To date, the existing research findings have not introduced comparative analysis for different locations in Egypt in terms of PV power generation prediction.

Performance analysis of a PV power plant can be considered a tool to determine the quality of the system. In this article, a 4.5 MWp grid-connected photovoltaic plant was real-time analyzed and simulated. This study aims the following:

1. Introduce a new approach to enhance the prediction quality by combining different solar models.
2. Select, monitor, and investigate the performance of the solar farm to ensure its reliability.
3. Validate the new approach by implementing the actual site data into solar performance modeling software to build an identical representative project to the field.
4. Using the new approach to predict the power generation of photovoltaic power plants for different locations in Egypt.
5. Study design parameters to suit the weather of Egypt, such as module types and tracking technology.

This study offers a methodology and decision-making support to help decision-makers choose a suitable solar power plant location based on power prediction for future projects in Egypt, study the performance of current projects, and help decrease the grid imbalance. Thus, the cost of grid management will be reduced. Furthermore, an accurate energy prediction would have a notable impact on preventing overloading and grant efficient energy storage.

The paper is organized as follows. Following the introduction, the methodology, system description, and validation are presented in Section 2; results and discussion are presented in Section 3. Finally, the conclusions are presented in Section 4.

2 | RESEARCH METHODOLOGY

To achieve the research objective of increasing the prediction quality, the research methodology is composed of three phases; phase one is the real-time monitoring and performance analysis of an actual solar field "Gold Tree Solar Farm" for the year 2019. The monitoring process collected data are from onsite weather station and sensors installed on arrays with intervals of 5 minute between each measure. These data are composed of the weather file and include the different solar angles, all types of irradiance coming from the Sun, the actual specification of the equipment, solar panels, inverters, and panel's orientation during the day. This phase includes

TABLE 1 Review on grid-connected PV systems assessment analysis

Location	System capacity	Module type	Monitored period	Outcomes	Refs.
Eastern Turkey	2.13 MW_p Installed grid-connected	Poly	1 year	It is concluded that the predicted energy injected into the grid is 3506.4 MWh/year and PR 83.4%.	19
India	10 MW_p Installed grid-connected	Multi	2 years	The performance of the system is found to be lower than land-based plants, and the degradation rate is found to be high due to climatic conditions.	20
Navrongo, Ghana	2.5 MW_p grid-connected	Poly	3 years	As a conclusion, they found the performance ratio to be 70.4%, with an estimated energy cost of \$0.2411/kWh and preventing 3852 metric tons of carbon dioxide from being emitted.	21
North of Portugal	15 kW_p grid-connected	Poly	2 years	The results showed a 21% higher electrical production for the estimated values compared to the actual.	22
Different locations in India	1 MW_p Simulated Using PVsyst	Mono, Poly, CdTe, CIS	1 year	The results showed that CIS module type gives the highest annual PR and energy yield 86.9 and 2471 MWh. The maximum output after 25 years of operation is produced by Poly module.	23
Different locations in the United Kingdom and India	3 kW_p Simulated Using PVsyst	Multi	1 year	The results showed that the amount of energy generated could be similar even the solar resources are different due to the assumption of the degradation rates and the uncertainty of the energy output.	24
Northeastern Brazil	2.2 kW_p grid-connected	Poly	11 months	The results showed that the system PR was 82.9% and the annual energy yield was 1685.5 kWh/kW_p. It was concluded that the northeast region of Brazil is relatively good for PV system performance.	25
India	190 kW_p Installed and simulated grid-connected	Poly	1 year	It is concluded that the measured and estimated energy yield is close with an uncertainty of 1.4%. It is found that the inclination of the panels needs to be optimized.	26
Crete, Greece	171.36 kW_p, grid-connected	Poly	1 year	It is concluded that the system injected 229 MWh/year with a PR of 67.36%	27
Dublin, Ireland	1.7 kW_p, grid-connected	Mono	1 year	The annual PR and capacity factor are 81.5% and 10.1%, respectively.	28
Southern, Spain	2 kW_p grid-connected	-	1 year	It is found that the PR of the system is 67.9% and the annual final yield is 1424 kWh/kWp.	29
Alexandria, Egypt	10 MW_p	Mono, Poly	1 year	The system performs with a good PR 84% and energy injected into the grid of 153.3 GWh/year	30

studying the correlation between five main parameters that are plant power, irradiance, wind speed, ambient temperature, and panel temperature. Furthermore, phase two includes the solar performance modeling software validation to predict the electrical power produced for any solar farm. It consists of validation and new approach implementation, which includes designing and building a representative identical model to the actual field using solar performance modeling software "PlantPredict" then optimizing this model to get the accurate estimated energy. After establishing the prediction model and optimizing it, the results will be compared with the actual power generation data collected from the field. Finally, the identical model is used as a pilot project in Egypt's case study in phase three; this phase consists of three main stages, which are choosing hotspots locations in Egypt using Solar Atlas, then predicting the energy of each location and studying different design parameters to determine which suits the weather file of Egypt. Moreover, this study offers a methodology for the decision-makers to be able to choose the suitable solar power plants location based on power prediction for any future projects in Egypt. Figure 1 shows the methodology phases in a flowchart.

2.1 | Gold Tree Solar Farm system description

Gold Tree Solar Farm is an 18.5-acre solar farm located at the California Polytechnic State University see Figure 2. The university is located in San Luis Obispo County in central California. The plant is located 2 miles northwest of the Campus with coordinates: 35.32/35°19′12″latitude and −120.69/−120°41′24″longitude.

To improve the power generation of the solar field, the Gold Tree Solar Farm uses a single-axis tracking system. The solar panels rotate around a north-south axis and follow the east-west movement of the Sun. A motor controlled by tracking software ensures that the panels are positioned to face the Sun. The facility consists of approximately 16 000 solar panels, which have a capacity of 4500 kW_{AC} or 5658 kW_{DC}. The field is divided into two main sections, each one using a 2500 kVA rated transformer from GC Power Systems USA. The system is ground-mounted since it is on a location, which has a large space away from buildings shading. The modules are mounted on a steel structure about 5 ft from the ground to have an easy access for maintenance and inspection. The panels depend on natural air circulation around the arrays to avoid overheating. As mentioned earlier, the plant uses two identical transformers that are connected

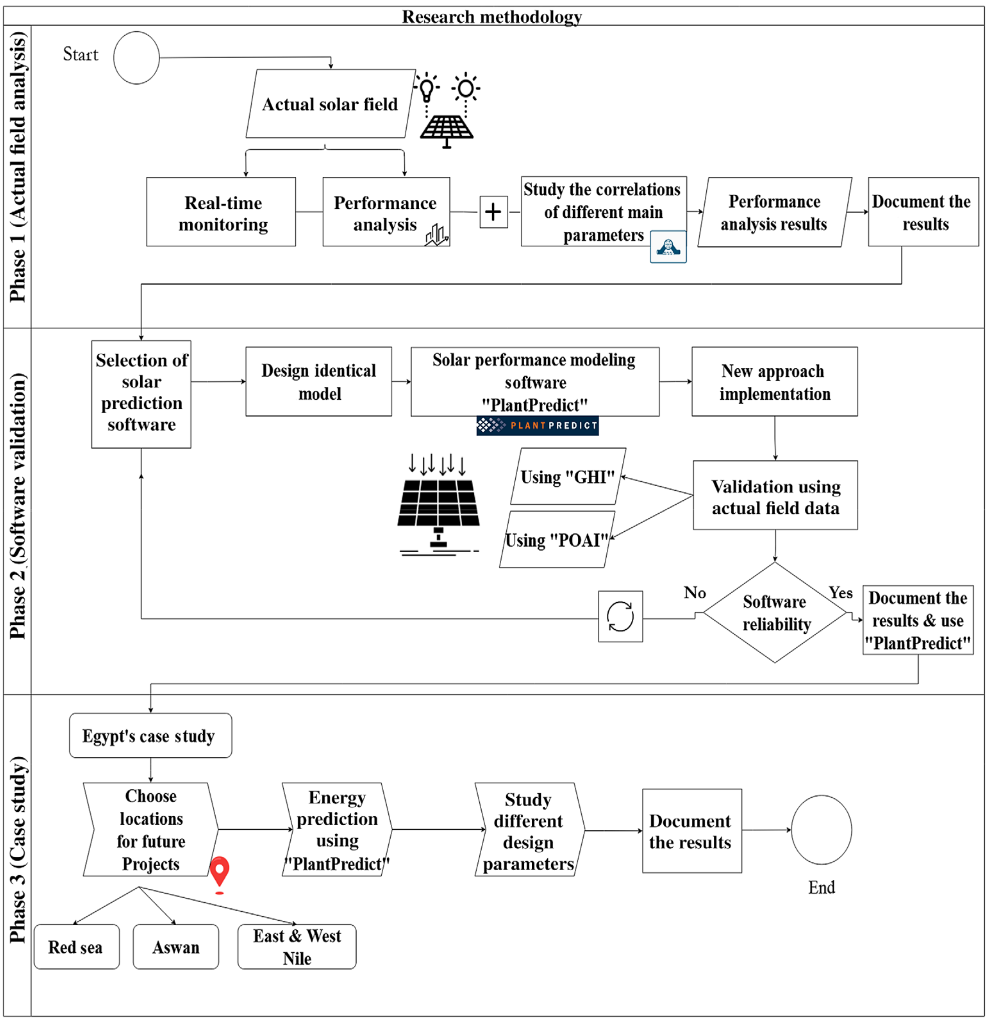


FIGURE 1 Research methodology diagram

to a total of 75 inverters. Of the 16 379 photovoltaic modules, 8664 are manufactured by Trinasolar and the other 7733 by REC Solar. All modules have a peak power of 345 W_p and are arranged in strings, with each string containing 19 modules in series. Each inverter can have up to 12 strings attached to it. The ground coverage ratio of the solar farm is 58.5%. The tracker range of motion is -52° to $+52^{\circ}$ with backtracking technology. The plant specifications are shown in Table 2.

2.2 | Building model validation

The solar performance modeling software is first validated in this work by using a real solar field. To accomplish this task, a simulation model is designed for the Gold Tree Solar Farm using "PlantPredict." The plant parameters and weather files like the temperature, POAI, GHI, and wind speed are monitored and extracted from the actual site. The number of simulations is developed for four different weeks in the months of January, April, July, and October 2019. These 4 months cover significant fluctuations in power generation and cover all four seasons.

FIGURE 2 Gold tree solar farm

TABLE 2 Plant specifications

DC capacity	5.7 MW
AC capacity	4.5 MW
DC/AC ratio	1.26
Module arrangement and quantity	16 379 total number of modules • 8664 modules (Trinasolar Manufactured) 459 strings with 19 modules each • 7733 modules (REC Solar Manufactured) 407 Strings with 19 modules each
Mounting	• Ground Mounting with single-axis tracker • Tracker: Array Technologies Inc. Duratrack HZ V3 Tracker range of Motion
Inverter type and quantity	Solectria Renewables PVI-60TL, 75 inverters
Row Spacing	3.35 m
GCR	0.59

Based on the weather files, two approaches are used to test the accuracy of the prediction. The first approach is to use the POAI data to determine irradiance on the panels, and the second approach is to use only the GHI data. That second approach requires manipulation of the irradiance data to determine the actual irradiance on the panels and consequently is not as precise as the first approach. However, for a prospective solar field, it is probably the only irradiance data available for the system designer. Table 3 provides a more detailed description of the models.

2.3 | Model optimization

This research includes a number of simulations where the POAI is available to simulate the solar irradiance on the panels. These simulations are used for the validation of the model as we already had the data from the actual farm in operation. These cases do not need any irradiance transposition and decomposition models for the simulation. For locations in Egypt, the GHI approach had to be used. Considering the model optimization, the best performing transposition and decomposition models for these cases are Perez and DIRINT models.

Modeling POAI from GHI involves two steps, see Figure 3:

1. Decomposition of GHI into DNI and DHI.
2. Transposition of these components to POA of the modules.

Reindl Model: It is a function that computes the DNI and DHI from the GHI according to the Reindl correlation model if no DNI or DHI is available. There are three versions of the model available in the literature. The first is a function of zenith angle, clearness index, air temperature, and relative humidity, then the function of zenith angle and clearness index, then the function of clearness index only.

Erbs Model: It is a routine that estimates the DNI and DHI using the Erbs correlation.

DIRNIT Model: This routine determines direct normal irradiance using the DIRINT modification of the DISC model. DISC model is a routine that uses the DISC correlation function, which is used by both the DIRINT decomposition and Hay transposition models.

TABLE 3 Environmental conditions data and simulation settings for the four seasons of 2019

Parameters	January	April	July	October
Location	San Luis Obispo, California, United States			
Elevation	144.26 m			
Wind sensor height	10 m			
Time zone	−8			
Time step	5 minutes			
Start-end date	January 1, 2019-December 31, 2019			
Start-end date for the visualized periods	06-12	14-21	07-14	06-12
Time shift	9 minutes	62 minutes	68 minutes	49 minutes
Time step	5 minutes			
Rotational limits	−52°/52°			
Array Azimuth	180°			
Soiling factor	2%			
Decomposition model	DIRINT			
Transposition model	PEREZ			
Direct shading	Linear			

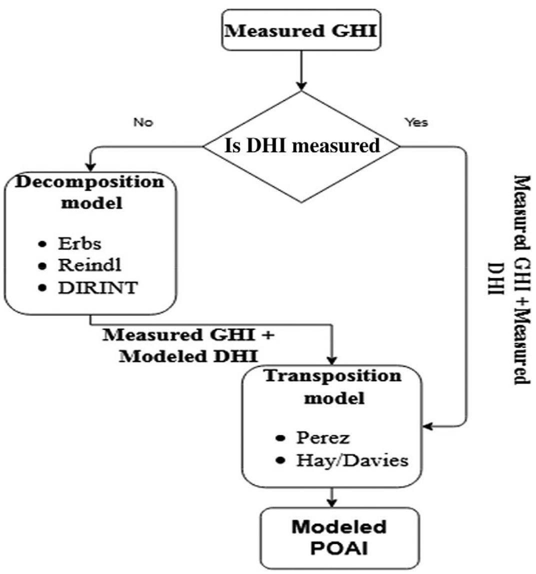


FIGURE 3 Modeling process and models considered

Perez or Hay Models: they are functions that geometrically transpose the DNI on an arbitrary tilted plane and transpose the DHI on this titled plane. The transposition and decomposition models considered are shown in Table 4[31-36]

2.4 | Modeling approach validation

The model optimization is needed when using GHI as the source of irradiance in the weather file. The combination between transposition and decomposition models showed that DIRNIT and Perez combination give the smallest relative error at 2.9%, as shown in Table 5 and Figure 4; this result is referred to the combination of extraterrestrial irradiance and the sky clearness as a function of the diffuse, direct normal irradiance, and zenith angle in addition to the brightness coefficients in Perez model. Figure 5 illustrates the predicted energy distribution from the transposition and decomposition models.

3 | RESULTS AND DISCUSSION

The results of the investigation are presented in three different sections representing each phase of the methodology. The first section shows the performance analysis of Gold Tree Solar Farm for the monitored year and the correlations between different parameters. The second section discusses the validation for the software model, while the third one discusses the energy potential for different locations in Egypt. Finally, the last section discusses the possibility of changing the tracking technology and/or module types to better suit the weather in Egypt.

3.1 | System performance analysis results

A performance analysis for the system is performed for the whole year of 2019, and to visualize the data,

TABLE 4 Transposition and decomposition models

Model	Equation	
Reindl	$\mathrm{DHI}=f_d\times\mathrm{GHI}$	(1)
Inputs: K_T, GHI, and θ_z	$\mathrm{DNI}=\frac{\mathrm{GHI}-\mathrm{DHI}}{\cos\theta_z}$	(2)
Outputs: DNI and DHI	1-If $0\le K_T\le 0.3$, $f_d=1.02-0.254K_T-0.0123\cos\theta_z$	(3)
	2-If $0.3\le K_T\le 0.83$, $f_d=1.4-1.749K_T+0.177\cos\theta_z$	(4)
	3-If $0.83\ge K_T$, $f_d=0.486K_T-0.182\cos\theta_z$	(5)
Erbs	1. Using Equations (1) and (2)	
Inputs: K_T, GHI, and θ_z	2. 1-If $0\le K_T\le 0.22$, $f_d=1-0.09K_T$	(6)
Outputs: DNI and DHI	3. 2-If $0.22\le K_T\le 0.8$, $f_d=0.9511=4.388K_T^3-16.638K_T^3+12.338K_T^4$	(7)
	4. 3-If $0.8<K_T$, $f_d=0.165$	(8)
DIRINT	$\mathrm{DNI}=G_oK_n\times\mathrm{DIRINT}_{\mathrm{coeff.}}$	(9)
Inputs:	$K_n=K_{nc}-\Delta K_n$	(10)
m_a, K_T, G_o, and θ_z	$K_{nc}=0.866-0.122m_{a,p}+0.0121m_{a,p}^2-0.000653m_{a,p}^3+0.000014m_{a,p}^4$	(11)
Outputs: DNI	To get $\mathrm{DIRINT}_{\mathrm{coeff.}}$ Use K'_T, $\Delta K'_T$ and θ_z, $\Delta K_n=ae^{cm_{a,p}}$	(12)
	$K'_T=\frac{K_T}{1.031e^{\frac{-1.4}{\left(0.9+\frac{9.4}{m_a}\right)+0.1}}}$	(13)
	$\Delta K'_T=0.5\left(\left\lvert K'_T-K'_{\mathrm{Tnext}}\right\rvert+\left\lvert K'_T-K'_{\mathrm{Tlast}}\right\rvert\right)$	(14)
Perez	$\mathrm{POAI}=\mathrm{POAI}_{\mathrm{Tilted}}+\mathrm{POAI}_{\mathrm{Diffused\ tilted}}+\mathrm{POAI}_{\mathrm{Reflected}}$	(15)
Inputs:	$\mathrm{POAI}_{\mathrm{Tilted}}=\mathrm{DNI}\cos\theta$	(16)
$m_{a,p}$, K_n, K_T, GHI, DHI, DNI, G_o,	$\mathrm{POAI}_{\mathrm{Reflected}}=\rho_{gnd}\frac{\mathrm{GHI}(1-\cos\beta)}{2}$	(17)
ρ_{gnd}, θ, and θ_z	$\mathrm{POAI}_{\mathrm{Diffused\ tilted}}=\mathrm{DHI}(1-F_1)\cos^2\frac{\beta}{2}+F_1r_B+F_2\sin\beta$	(18)
Outputs: POAI	$r_B=\frac{a_o}{a_1}$, $a_o=\max(0,\cos\theta)$, $a_1=\max(\cos 85,\cos\theta_z)$	(19) (20) (21)
	$F_1=\max\left\{0,\left(F_{11}+F_{12}\Delta+\frac{\pi}{180}\theta_zF_{13}\right)\right\}$, $F_2=F_{21}+F_{22}\Delta+\frac{\pi}{180}\theta_zF_{23}$	(22), (23)
	To compute the brightness coefficients F_1 & F_2use ε bin	
	$\varepsilon=\frac{\frac{\mathrm{DHI}+\mathrm{DNI}}{\mathrm{GHI}}+5.535\times10^{-6}\theta_z^3}{1+5.535\times10^{-6}\theta_z^3}$, $\Delta=m_{a,p}\frac{\mathrm{DHI}}{G_o}$	(24), (25)
Hay/Davies	Using Equations (15), (16) & (17)	
Inputs:	$\mathrm{POAI}_{\mathrm{Diffused\ tilted}}=\mathrm{DHI}\left[K_n\frac{\cos\theta}{\cos\theta_z}+(1-K_n)\cos^2\left(\frac{\beta}{2}\right)\right]$	(26)
$m_{a,p}$, K_n, K_T, GHI, DHI, DNI, G_o,		
ρ_{gnd}, θ, and θ_z		
Outputs: POAI		

different periods covering the four seasons for five main parameters, plant power, irradiance, wind speed, ambient temperature, panel temperature are shown in the results.

The overall PR, annual capacity factor, and energy injected into the grid in 2019 are 79.2%, 27.7%, and 10.92 GWh/year, the monthly energy distribution, GHI, and the ambient temperature are represented in Figure 6. Figure 7 shows the accumulated monthly PR and energy production. It is observed that the PR in January is 92% with energy production of 426 095 kWh, which is the lowest energy production, while in July, the accumulated energy reached its maximum value of 1 175 275 kWh with the lowest accumulated PR of 71%. Furthermore, 80% and 77% accumulated PR for April and October, respectively.

The variation of the parameters for the monitored period is illustrated in Figure 8, showing that the system unsurprisingly is strongly dependent on the irradiance parameter with the highest correlation coefficient above 0.9 in all seasons. It is found that there is a strong positive direct correlation between power and the panel temperature, with correlation coefficient reaching 0.95. The correlation between power and ambient temperature is observed to be strong. It is shown that the wind speed correlates strongly with the panel temperature; in a simple word, an increase in the wind speed would help to balance the panel temperature that would increase the energy produced. However, the wind speed does not affect the power directly, but it has a strong direct effect on the panel temperature that directly affects the power. All the correlation increase in hot weather conditions except the panel temperature with the power it decreases as shown in Table 6. Figure 9 shows hourly PR for the site in the four seasons, the data show that the highest PR at the peak Sun hour is in January with a performance ratio over 95%, while the lowest PR occurs in July below 75%, April, and October has a moderate PR between 75% and 85%. It is obvious that in summer as the ambient temperature reaches its maximum value leading to an increase in the panel temperature as well as

TABLE 5 Comparative case study of different models combination relative errors

Combined Models	Actual Energy	Erbs and Perez	DIRINT and Perez	DIRINT and Hay	Reindl and Perez	Reindl and Hay	Erbs and Hay
Relative Error%	0	3.73	2.9	4.2	3.75	5.12	5.1

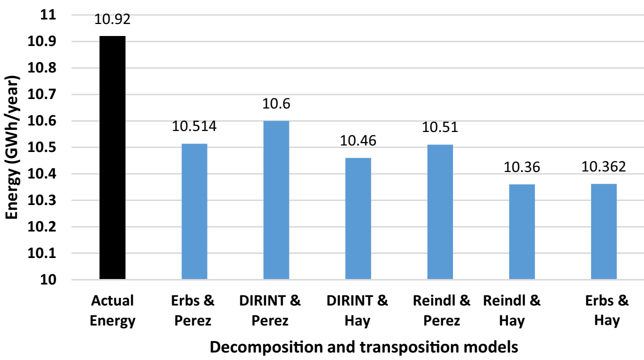


FIGURE 4 Comparison of energy injected into the grid by transposition and decomposition models

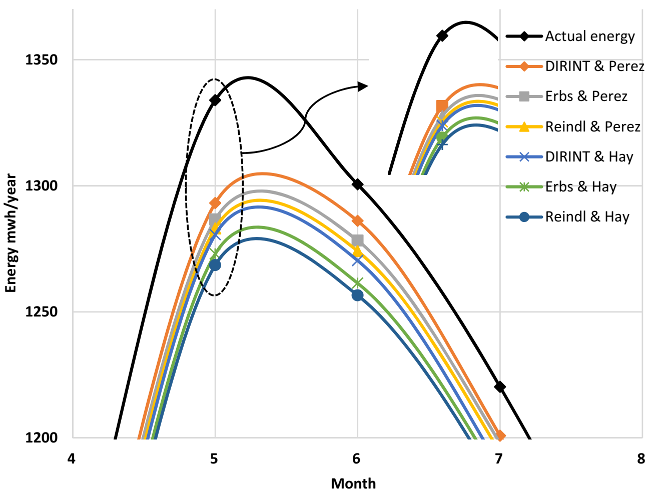


FIGURE 5 Transposition and decomposition predicted energy distribution

the low wind speed, considering the previous correlations. Thus, the system performs with the lowest PR. It appears that the site performs with an acceptable range of PR and even more in cold weather.

3.2 | Simulation results

For these simulations, the system design, geometrical data, module parameters, and inverter characteristics are incorporated into the "PlantPredict" software. The study is conducted for the whole year of 2019, and to visualize the data in different periods in the year. The results presented here show the study covering four different months with 5-minute intervals. One week in each month is carefully selected to show the time where weather fluctuations occur. These simulations are performed using the POAI data first and then the GHI data. The simulated outcomes have good agreement with the actual measured data. The overall energy predicted for 2019 is 10.86 GWh/year and 0.54% relative error when using POAI and 10.60 GWh/year and 2.9% relative error. The RMSE and MAE using the POAI are 0.042 and 0.06, respectively, while using GHI is 0.226 and 0.32, respectively.

The values of the POA insolation and PR of the PV plant are shown in Tables 7 and 8. Using the POAI approach, the simulations showed the highest total monthly POA insolation of 61.34 kWh/m^2 in July for the solar PV plant. Using the GHI approach, it exhibited a similar total monthly POA insolation of 63.61 kWh/m^2 in July. The PR of the plant is around 78% to 80% during the monitored period with the POAI approach, whereas the GHI approach resulted in a variable PR of 71.69% to 80.09%. The brief comparison throughout the four selected months between different performance parameters from the simulation study is highlighted in Tables 7 and 8. Considering the different seasons, PR presents a minimum value of 79.72% and a maximum of 79.94% during winter. Instead, it changes between 72% and 78% in summer. This result is due to the higher average ambient temperature in summer, which results in higher module temperatures and consequently lower panel performance.

Figure 10 illustrates the energy generation predicted by the PlantPredict software compared with the actual energy generated by the farm for a period between January 6 and January 12, 2019. This chosen period represents typical data for the winter in California. The peak energy generation of 4 MWh is observed from the 12 January data, as it is the clearest day with maximum POAI while the minimum energy is on the fourth day of the week when the peak energy generated is only approximately 0.8 MWh. That low peak of energy generated on 9 January is due to rain and cloudy weather, which resulted in low irradiance.

Using the POAI approach in Figure 10A, the trend shows an underestimation of the actual power for the

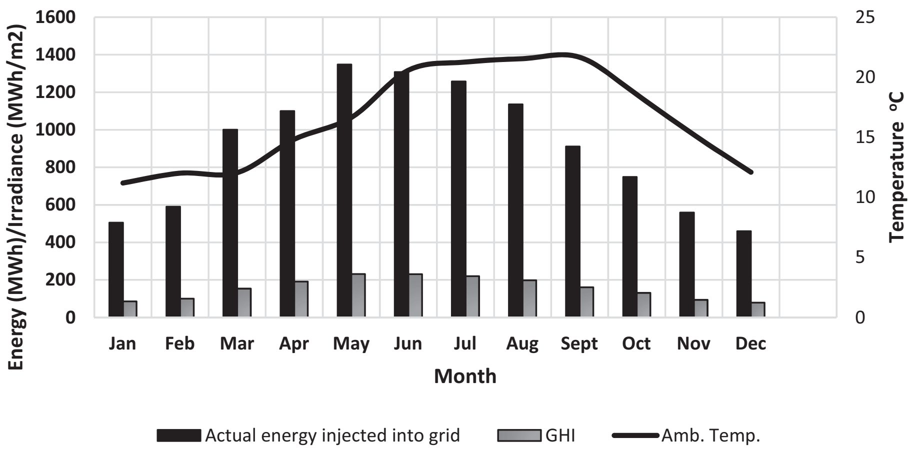


FIGURE 6 Monthly Energy injected into the grid, GHI, and the ambient temperature

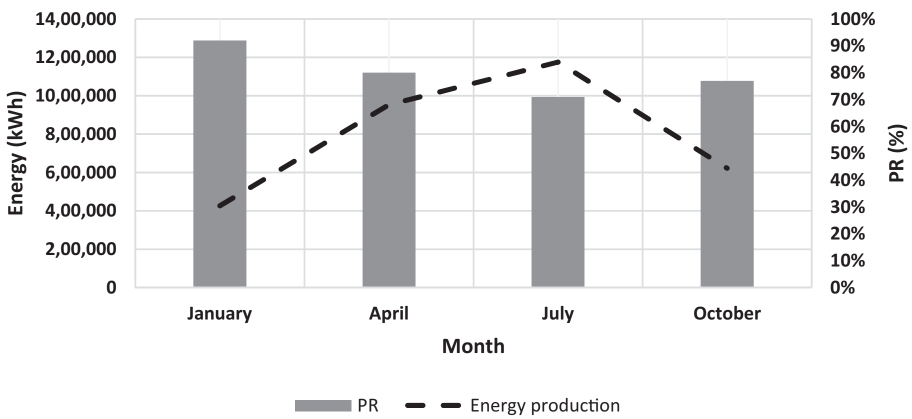


FIGURE 7 Accumulated monthly PR and energy production

whole period with a total actual energy generation of 75.7 MWh and total predicted energy generation of 73 MWh with a relative error of 3.6%. While using the GHI approach in Figure 10B, the same trend is observed for distributing actual and predicted energy. It can be seen that there is a power generation underestimation over most of the period and an overestimation at the time of power peak. The trend also shows an underestimation for the rest of each day as the Sun disappears. By comparing the total actual energy with the total energy predicted using the GHI approach of 81 MWh, the relative error is 6.9%. As it can be seen here, the energy predicted using the GHI approach has higher error than the POAI because POAI data include exact measurements of the irradiance on the panels compared with the GHI approach that uses models to infer that information.

Figure 11 shows the same trend for distributing generated energy predicted and compares it with the actual measured energy generation for a period between April 14 and 20, 2019. This trend represents typical data for the spring in California. An approximate maximum value of energy generation of around 4.3 MWh is reached on a number of days except 20 April. On that day, the maximum energy reached is 3.2 MWh as it is a cloudy day with high irradiance fluctuations. The spring in California has a medium irradiance level resulting in moderate energy generation.

Using the POAI approach in Figure 11A, a slight underestimation in the predicted energy generation is observed at the beginning of the day, followed by an overestimation before the middle of the day. An underestimation is also observed at the peak of power generation, which is followed by a rapid decrease with the Sun disappearance. On the other hand, using the GHI approach, as shown in Figure 11B, the behavior indicates an underestimation at the beginning of every day, then a sudden increase at the peak compared with the actual energy generated. Again here, it is followed by a rapid decrease with the Sun's disappearance. The total actual generated energy of 235.81 MWh is recorded, and the total predicted energy generated using the POAI approach is found to be 222 MWh with a relative error of 5.85%. Using the GHI approach, the predicted energy generated is found to be 216 MWh with a relative error of 8.4%.

Figure 12 illustrates the energy generation distribution predicted by the “PlantPredict” software compared with the actual energy generated by the farm for a

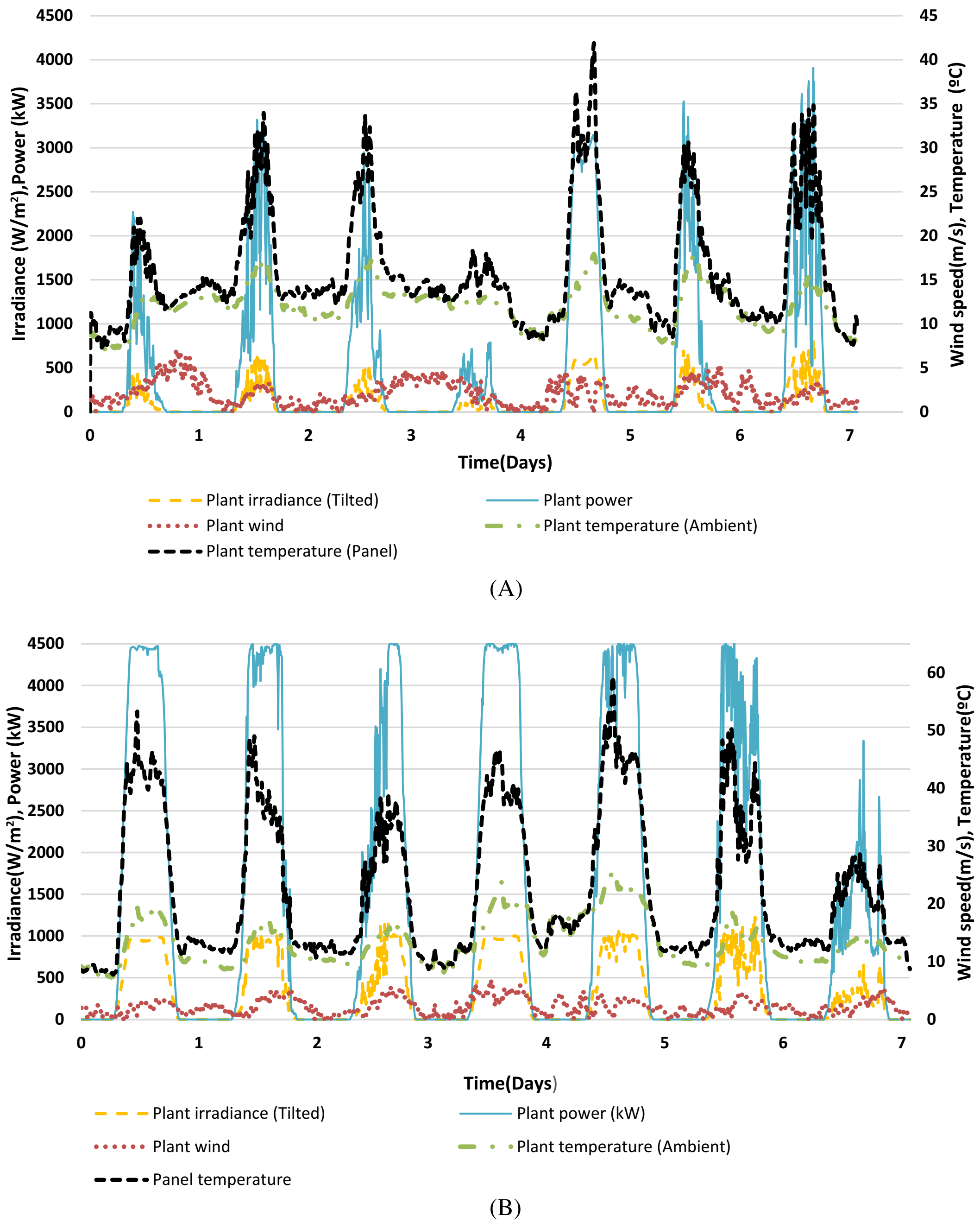
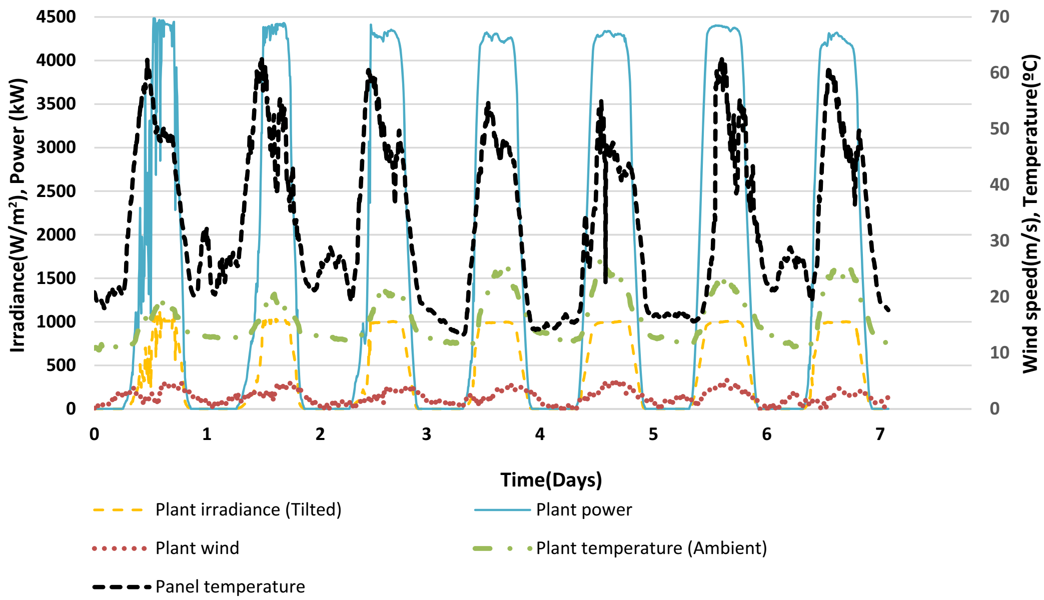


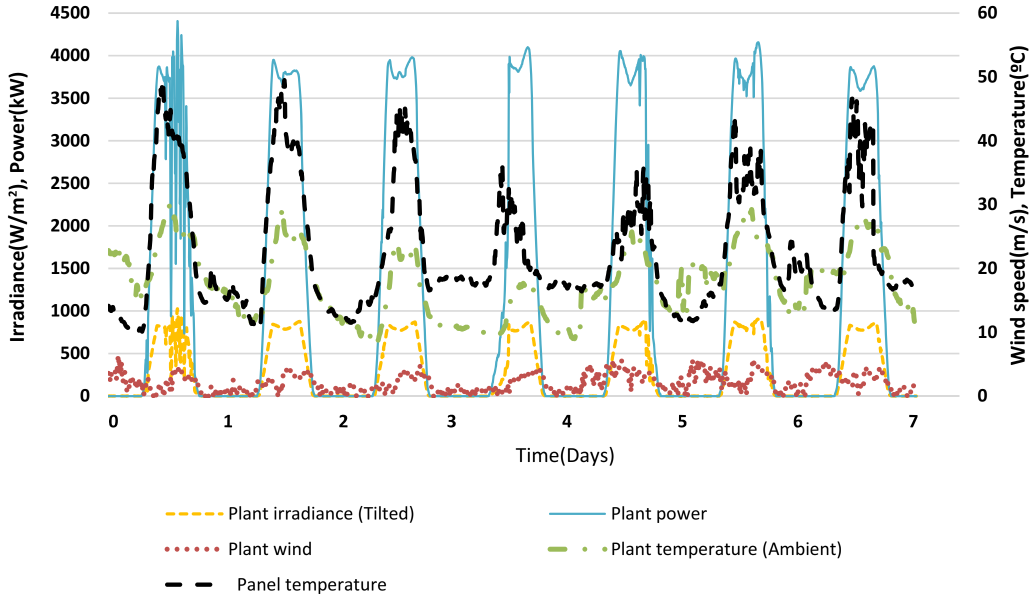
FIGURE 8 Plant irradiance, pant power, wind speed, ambient and panel temperature for the monitored period A, 6-12 January; B, 14-21 April; C, 7-13 July; D, 6-12 October

period between July 7 and 17, 2019. This period in July represents typical data for the summer months in California. The actual generated energy, reaching its maximum capacity of 4.5 MWh, is nearly the same for each day of that week with only slight daily differences. The summer has the highest energy generation over all other seasons due to its high irradiance and long daytime duration. The absence of cloud coverage also contributes to the high power generation during those months of summer.

Using the POAI approach and GHI approach, as shown in Figure 12A,B, a slight underestimation for the predicted energy occurs in the early part of the day. That underestimation slowly moves to an overestimation in the middle of the day. The energy generation output during the monitored period is measured to be 273 MWh, and the total predicted energy using the POAI approach is found to be 271 MWh with a relative error of 0.78%. The total predicted energy generation using the GHI approach is found to be slightly



(C)



(D)

FIGURE 8 (Continued)

lower at 258 MWh, which represents a relative error of 5.5%.

The energy generation distribution of the predicted and the actual energy generated by the farm for a period between 6 and 12 October, 2019 is shown in Figure 13. The maximum energy generation is recorded on October 6 at approximately 4.3 MWh. For the rest of the week, the actual peak energy generation varies between 3.8 and 4.1 MWh. Because of cloudy and unstable weather, autumn in California provides a lower irradiance and lower overall energy generation potential.

Using the POAI approach in Figure 13A, a mixture of a slight underestimation and overestimation can be seen for the predicted energy at the beginning of the day. At the peak power generation time, the model overestimates the actual generation, then it slightly underestimates at the end of the day. The total actual energy of 205.4 MWh and total predicted energy of 194 MWh represent a relative error of 5.5%, while using the GHI approach in Figure 13B, the trend shows an underestimation of energy generation at the beginning of the day then an overestimation during the peak period until it goes down

TABLE 6 Correlation coefficients for the parameters within the monitored period

Monitored period	Case	Correlation coefficient (r)
January 6-12	Plant Power and Irradiance	0.9974
	Plant Power and Ambient Temperature	0.6066
	Plant Power and Wind Speed	0.108
	Plant Power and Panel Temperature	0.9245
April 14-21	Plant Power and Irradiance	0.9946
	Plant Power and Ambient Temperature	0.7237
	Plant Power and Wind Speed	0.54327
	Plant Power and Panel Temperature	0.958
July 7-13	Plant Power and Irradiance	0.9972
	Plant Power and Ambient Temperature	0.8977
	Plant Power and Wind Speed	0.642
	Plant Power and Panel Temperature	0.7994
October 6-12	Plant Power and Irradiance	0.9966
	Plant Power and Ambient Temperature	0.6915
	Plant Power and Wind Speed	0.3768
	Plant Power and Panel Temperature	0.8545

similarly to the actual data at the end of the day. The total actual energy generation of 205.4 MWh compared with a total predicted energy generation of 195 MWh represents a relative error of 5%.

The actual total energy generation produced in a week at the location and the predicted generation using the POAI and GHI approaches are presented for the visualized four seasons as shown in Figure 14 and for the annual trend consult Figure 15. It can be observed that the maximum energy is produced in the summer as the highest solar irradiance occurs at that time of the year. Examination of the results shows the maximum energy generated at the solar farm is attained in July, with a value of 273 MWh, while the minimum potential is measured in January, with a value of 75.74 MWh. In "PlantPredict" simulation, using the POAI approach, maximum energy generated is predicted in July (271 MWh), and the minimum potential is estimated to be in January (73 MWh). In "PlantPredict" simulation, using the GHI approach, the maximum energy output is 258 MWh achieved in July. The minimum potential of 81 MWh is projected in January. The recorded data are compared with the simulated generation values. Using POAI and GHI, the relative error ranges are 0.78% to 5.8% and 5% to 6.9%, respectively. It can be observed that the error is relatively proportional to the increase in temperature, which can be referred to the module temperature loss calculation by the simulation tool.

3.3 | Power prediction for future projects in Egypt

Egypt is a solar belt country with a large amount of solar energy potential that generates numerous locations well suited to install solar farms. Using Solar Atlas,[1] the number of regions in the country is identified to be investigated. The chosen locations inside Egypt for the simulation models are Benban, Red Sea, West, and East Nile. See Table 9 for the geographical data.

Figure 16 shows the energy potential for a full year at the three chosen locations. The dataset used for the analysis hourly reported information for the entire year. The Red Sea region shows the highest energy potential with 14.39 GWh/year, while Aswan and West and East Nile regions have 13.88 and 13.80 GWh/year, respectively. Red Sea region also has the highest energy production for each month of the year because of the high wind speed at this location. This high wind improves the efficiency of the solar panels due to a natural cooling effect. That location also has the highest GHI of the three locations. Between January and March, Aswan has a higher energy generation trend than the West and East zone. From March to June, the energy generation seems to be reaching a plateau. From June to September, West and East Nile zone shows a higher energy generation potential than Aswan. Finally, from September to November, Aswan shows a higher energy generation potential again. The performance ratio for Aswan, Red Sea, and West and East Nile are 73.7%, 73.2%, and 74.8%, respectively. See Table 10 for a summary of the data. According to this analysis, the Red Sea is the best place for installing a PV power plant in Egypt due to the effect of the average ambient temperature and the maximum air temperature in the weather file of the location, which is relatively low in this location will enhance the system performance in the hot day of the year,[37] as well as the high average wind

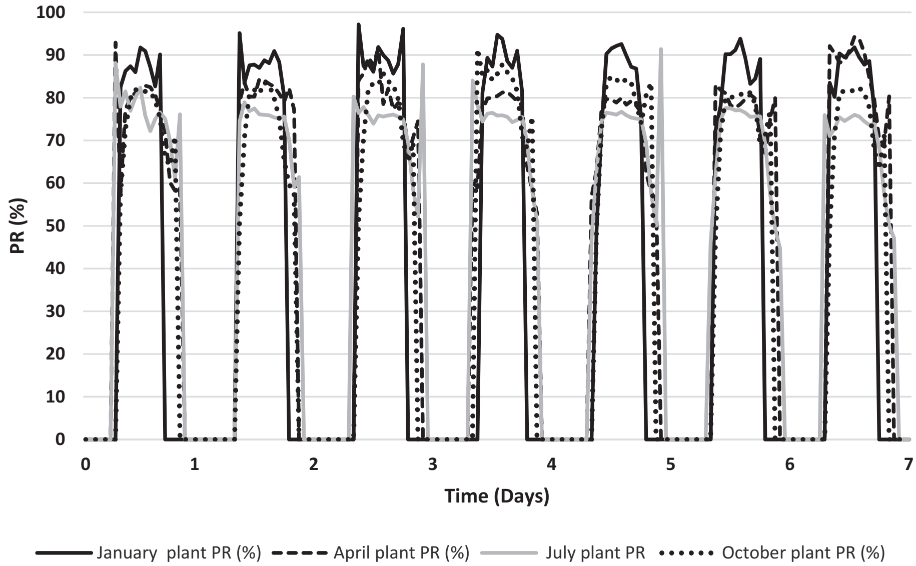


FIGURE 9 PR for the monitored periods

TABLE 7 Results of "PlantPredict "using POAI for the Gold Tree Solar Farm model

Parameters	January	April	July	October
Plant net energy MWh	73	222	271	205.37
POA insolation kWh/m^2	16.10	49.47	61.34	42.91
PR%	79.72	79.29	78.01	79.94

TABLE 8 Results of "PlantPredict "using GHI

Parameters	January	April	July	October
Plant net energy MWh	80.8	215	258	195
POA insolation kWh/m^2	17.84	52.82	63.61	45.03
PR%	80.09	72.28	71.69	76.54

speed in the Red Sea region rather than other locations, which would help in cooling the PV module and increasing the efficiency and performance of the system.[38,39] The estimated losses parameters for the three locations are shown in Table 11.

The difference in the total energy predicted is 0.5 GWh/year higher for Red Sea would relatively have an effect on the cost-saving as shown below:

$$0.5\frac{\text{GWh}}{\text{year}} = 500000\frac{\text{kWh}}{\text{year}} \times \text{cost of electric}\frac{\text{Energy}}{\text{kW}}$$
$$= 500000 \times 1.35 = 675000\,\text{Egyptian Pound}$$
$$= \$42187.5$$

The cost of electric energy used for the household segmentation price for the Egyptian government is 1.35 Egyptian Pounds.[3]

By using emission factor from electricity generation of 0.429 kg CO_2/kWh from "the EPA's eGRID emission factors based on 2018 data published in 2020."

The total energy of the Red Sea prediction model is as follows:

$$14.4\frac{\text{GWh}}{\text{year}} = 14400 \times 10^3\,\frac{\text{kWh}}{\text{year}} = 14400 \times \frac{10^3}{8760}$$
$$= 1643.8\,\text{kW}$$

$$1643.8\,\text{kW} \times 0.429 = 705.19\,\text{kg of } CO_2$$

This is a result of a small pilot project of around 18.5 acres taking into consideration that Egypt has been devoted huge number of lands for renewable energy resources, especially for solar energy, which would cover a large-scale project in the future; the table below shows the devoted land's size for the three locations[1]; theoretically, the 18.5 acres of the pilot model is equivalent to just only 0.23% of the 8095.238 acres of Hurghada location 1 (Al-Ahyaa), which is 99.77% larger so that the calculated cost would be 675000*99.77 = 6734*10^4 Egyptian Pounds ($4.2 Million) and the calculated emissions would be 705.19*99.6 = 70 236.9 kg of CO_2,while location 2 (Kilo-10) is 595 238 acres, which represents 99.995% larger than the pilot model so that the calculated cost would be much higher and the calculated emissions reduction would be higher. Table 12 shows the devoted lands from the Egyptian government.

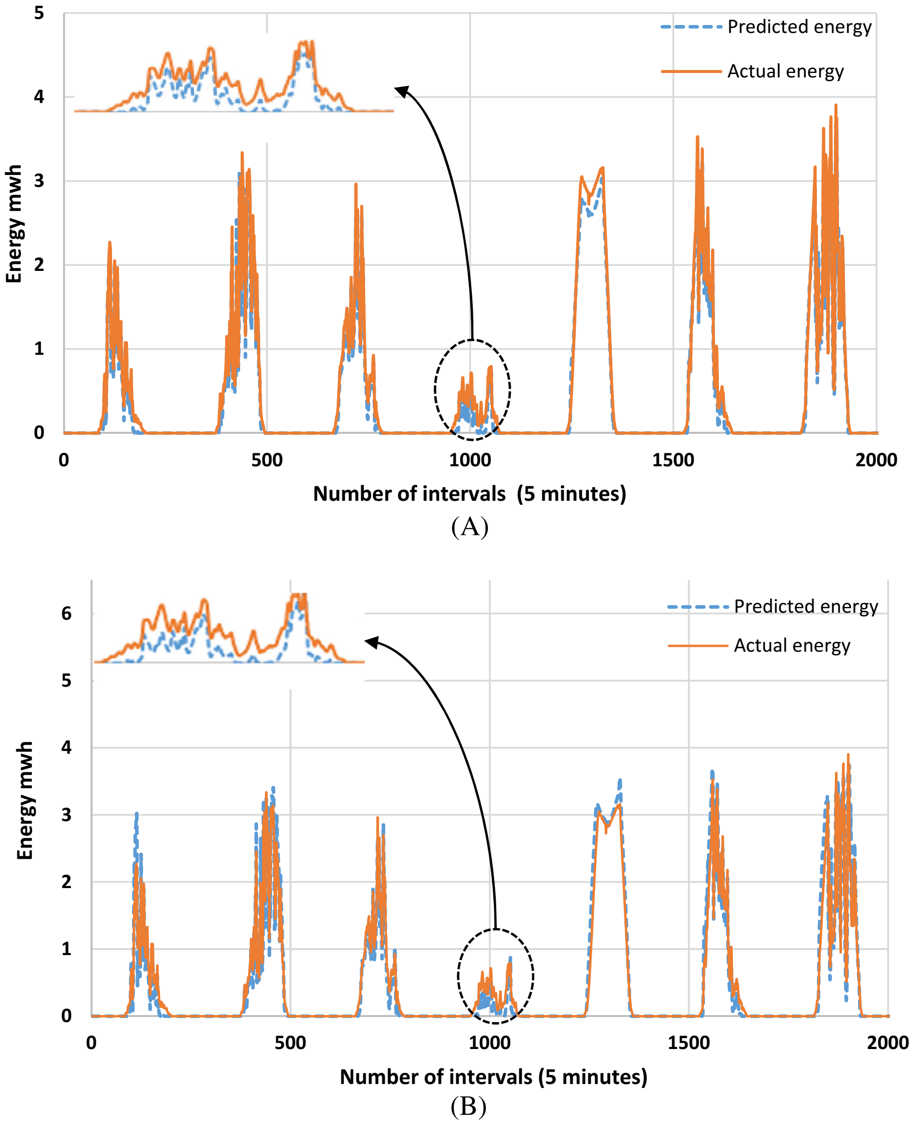


FIGURE 10 Energy distribution for 6 to 12 January, A, Using the POAI; B, Using GHI

3.3.1 | Design parameters analysis for Egypt's case study results

The model used for Egypt's location is chosen to investigate the changing of some design parameters of the solar field. The parameters we are mostly interested in investigating are the mounting technology and the module type. All the data presented here for these changing parameters are for the whole year with 60-minute intervals.

Figure 17 shows the monthly average distribution of GHI and ambient temperature for the whole year. The trend shows the lowest average GHI and temperature around 200 W/m^2 and 15°C, respectively, in January, which is similar to December. The highest average GHI occurs in June at around 350 W/m^2 and the highest average temperature around 32°C in July. From April to August, the average GHI reaches between 250 and 350 W/m^2, while the average temperature is between 25 and 35°C. It can be observed that there is no direct correlation between the GHI and ambient temperature.

Figure 18 shows the energy generation potential using two different solar tracking technologies. One technology is a fixed mount system, and the other one is a single-axis tracking system with backtracking. The results show that the use of single-axis tracking produces a 1.64 GWh/year higher energy generation, resulting in an increase in percentage of approximately 13.4%. In the case of monocrystalline and polycrystalline solar cells, we reached the same conclusion that the single-axis tracking increases the amount of daily solar radiation incidence[40] (see Figures 19 and 20). Consult Table 13 for more detailed comparison results and percentage of increase. All the data shown here are for an entire year, with each data point at a 60-minute interval.

Figures 19 and 20 show the effect of changing the mounting technologies and the module type. Figure 19A shows the difference between using Monocrystalline PV modules with fixed mount or single-axis tracking technologies. The trends clearly show that the use of single-axis tracking produces higher energy production. The

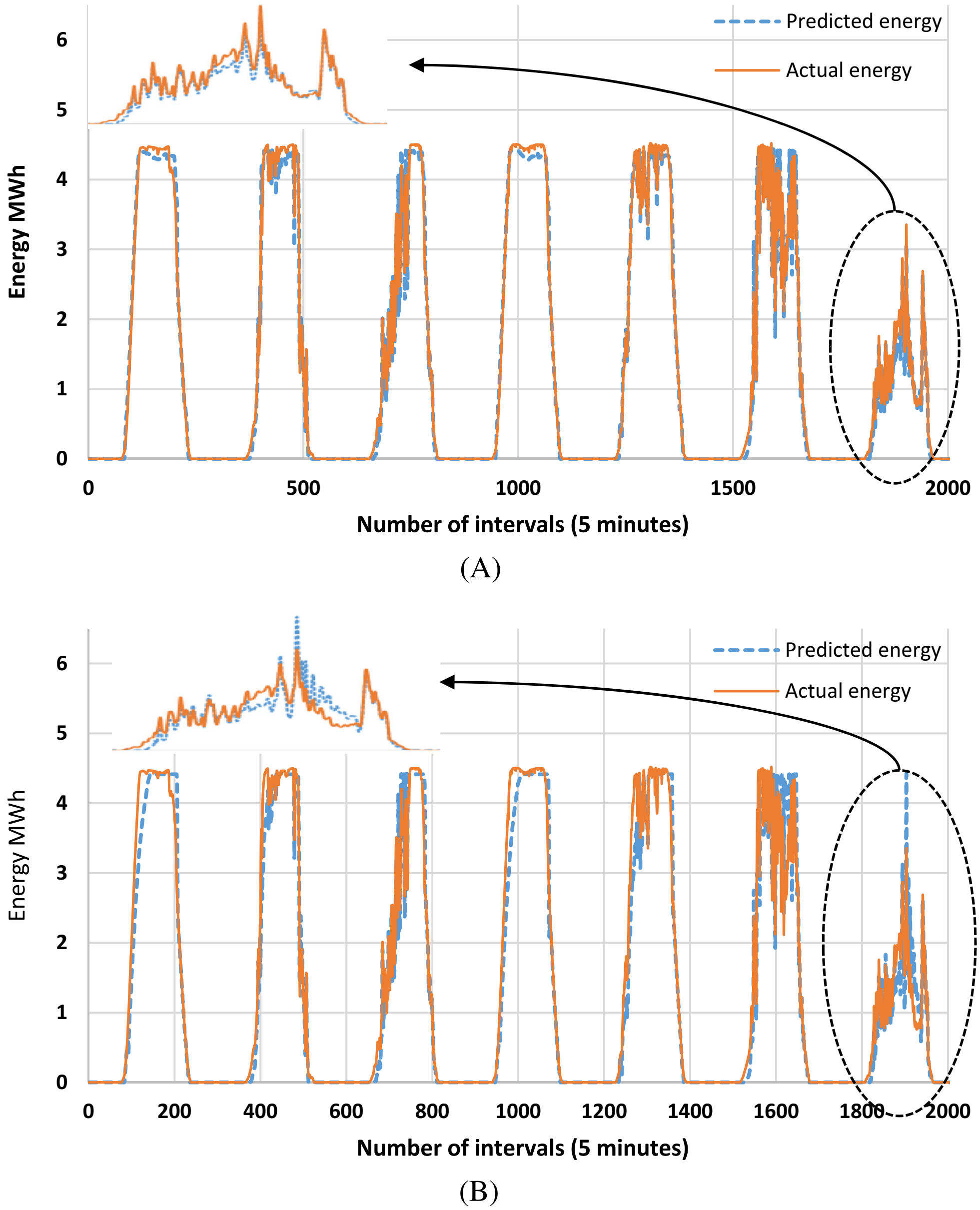


FIGURE 11 Distribution for 14 to 20 April, A, Using POAI; B, Using GHI

increase in the percentage of energy generation is 13.4%, which represents an energy difference of 1.64 GWh/year. See Table 14 for more detailed results.

Figure 19B shows the difference between using polycrystalline PV modules with fixed mount or single-axis tracking technologies. The trends show the usage of a single-axis tracking system generates an increased energy production of 13.26% and an approximate energy difference of 1.63 GWh/year. More results are shown in Table 15.

Figure 20A shows the comparison between the usage of two different types of modules with single-axis tracking technology. The first type is monocrystalline cells, while the second one is polycrystalline cells. The trend shows that they are nearly identical in energy production for this location. The percentage of energy generation increase between poly and mono is 0.5% (as shown in Table 16). The difference between the energy produced per year is approximately 0.07 GWh/year. This small difference is due to the monocrystalline efficiency decreases with the increase of the irradiance and ambient temperature, while polycrystalline shows a reverse behavior (see Figure 20B).[41]

Figure 20B shows the comparison between the two different types of modules using a fixed mount technology. The first type is monocrystalline, while the second one is polycrystalline. The trends again here show that they are mostly identical in energy production for this location. The percentage of energy increase between poly and mono is 0.5%, and the difference between the energy produced per year is approximately 0.08 GWh/year (as shown in Table 17). All in all, for the cases investigated in Egypt, the polycrystalline is a suitable choice. That choice would lead to a decrease in expenses for the PV farm installation as the polycrystalline module is cheaper than the monocrystalline module.

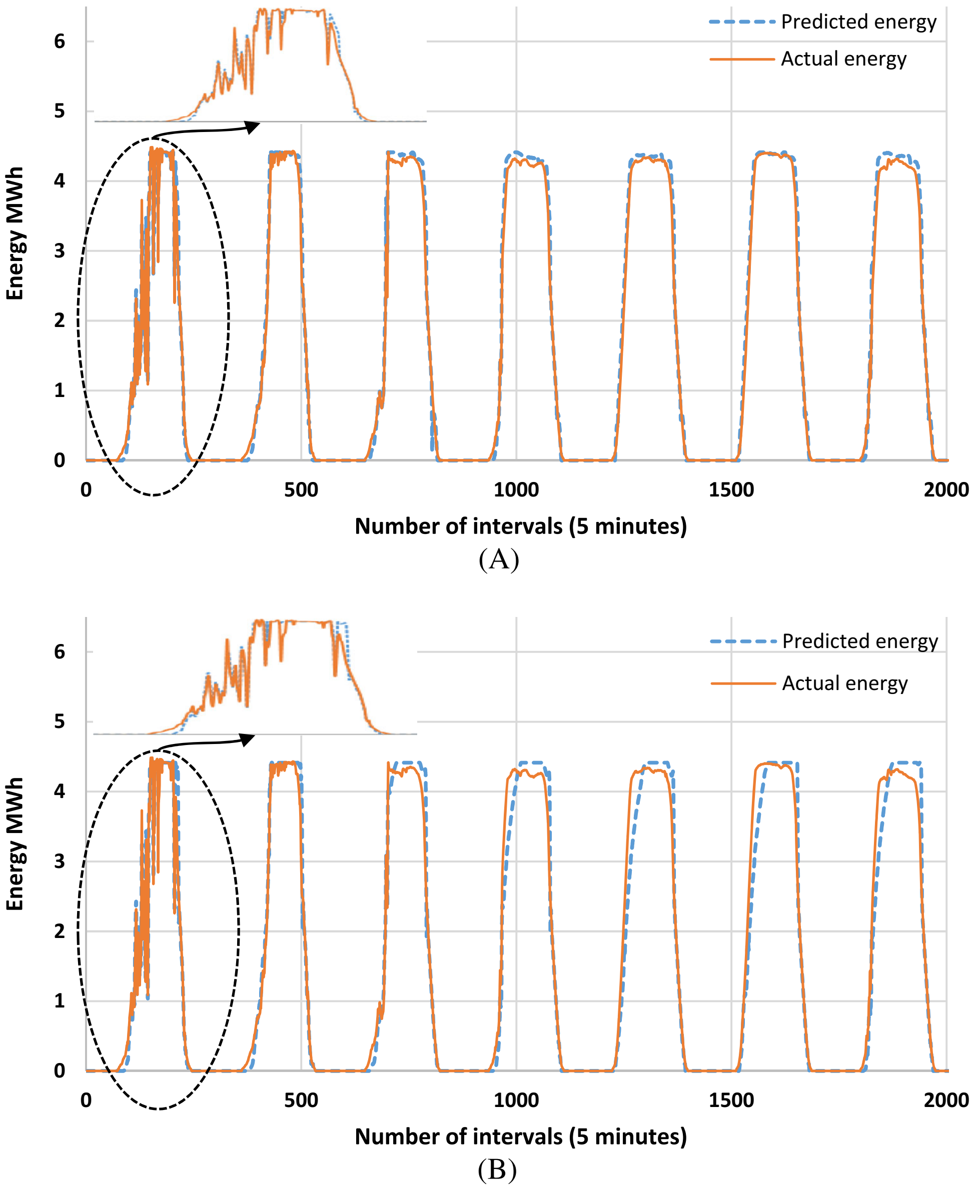


FIGURE 12 Energy distribution for 17 to 13 July, A, Using POAI; B, Using GHI

The difference in energy between using the polycrystalline is approximately 0.5% increase over the monocrystalline modules.

$$
\begin{aligned}
0.07\frac{\text{GWh}}{\text{year}} &= 70000\frac{\text{kWh}}{\text{year}} \times \text{cost of the electric}\frac{\text{Energy}}{\text{kW}} \\
&= 70000 \times 1.35 = 94500\,\text{Egyptian Pound} \\
&= \$5906.25
\end{aligned}
$$

For the emissions:

$$
0.07\frac{\text{GWh}}{\text{year}} = 70000\frac{\text{kWh}}{\text{year}} = \frac{70000}{8760} = 7.99\text{kW}
$$

$$
7.99 \times 0.429 = 3.42\,\text{kg}\,CO_2
$$

Theoretically, for a larger scale using real devoted land such as Al-Ahyaa location, the cost would be $94500 \times 99.6 = 9412200$ Egyptian Pound and the emission reduction would be $3.42 \times 99.6 = 340.6$ kg CO_2

Polycrystalline module is more feasible for the weather conditions of Egypt as monocrystalline module is sensitive to ambient temperature, shading, snow, dirt, while the polycrystalline module has less sensitivity to the ambient temperature. The monocrystalline behavior in hot weather months showed a gradual decrease in performance, while polycrystalline showed approximately the same behavior even on hot days.

3.3.2 | Energy loss factors for Egypt

The estimated loss parameters for the system when using fixed mounting are shown in Table 18, while when using tracking technology are shown in Table 19. One of the main loss parameters is the module temperature loss,

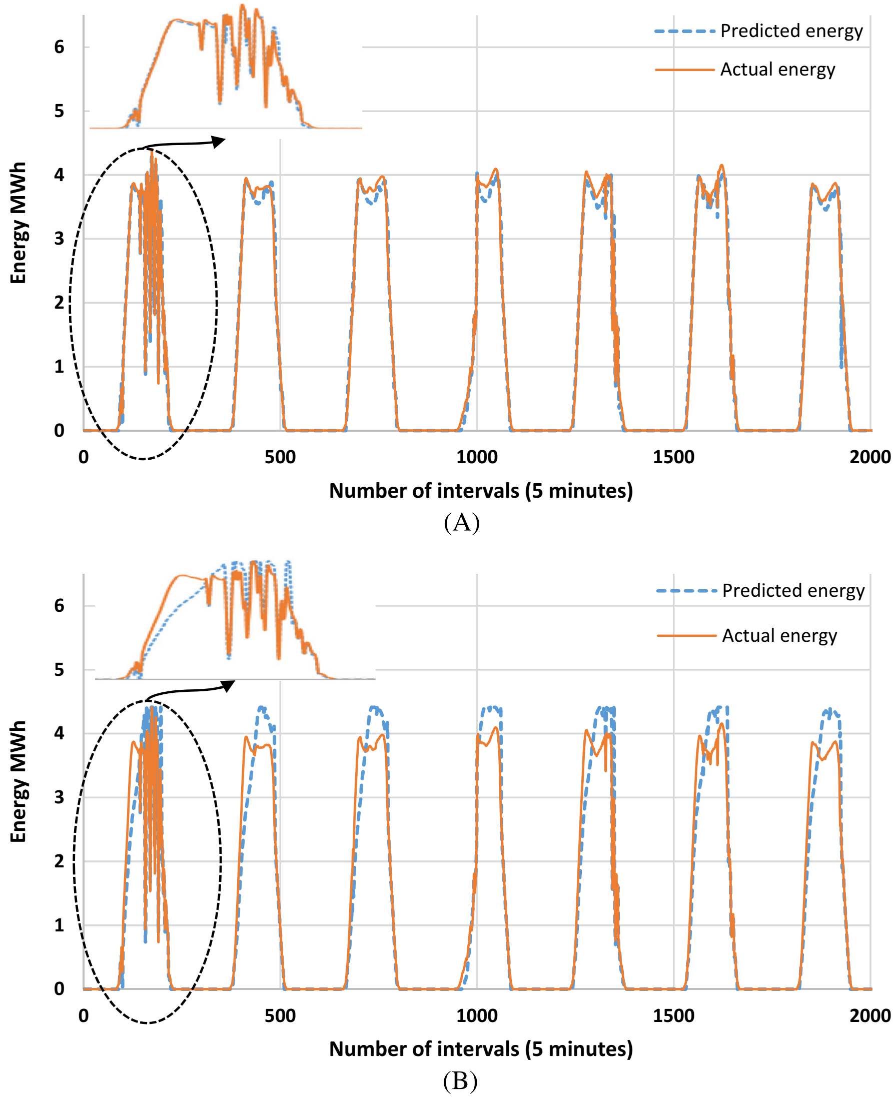


FIGURE 13 Energy distribution for 6 to 12 October, A, Using POAI; B, Using GHI

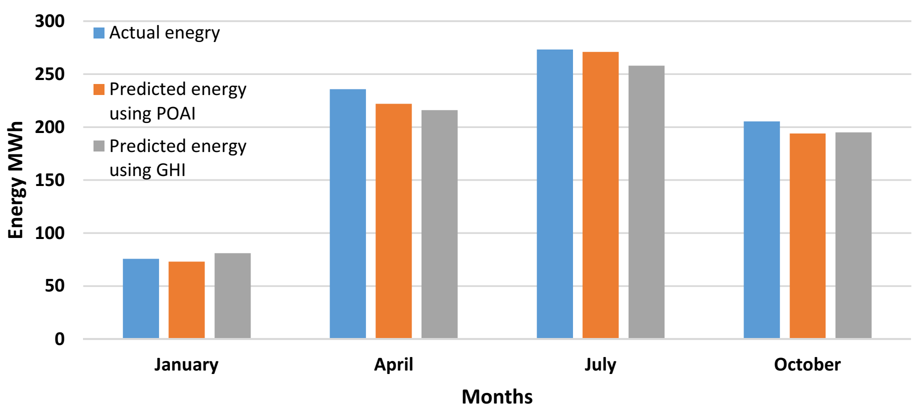


FIGURE 14 Monthly actual and predicted energy output

which can affect the whole system performance extremely. It is concluded that the annual module temperature loss when using polycrystalline technology ranges from 8.47% to 9.16%. In contrast, the loss due to monocrystalline technology ranges from 9.04% to 9.78%. These ranges contribute to increasing the superiority of the polycrystalline selection module over monocrystalline for Egypt's case study. IAM "Incidence Angle Modifier" factor on global correlate to the irradiance that does not reach the cell surface due to the incidence angle. It can be observed that the IAM factor is lower when using tracking technology compared with fixed mounting.

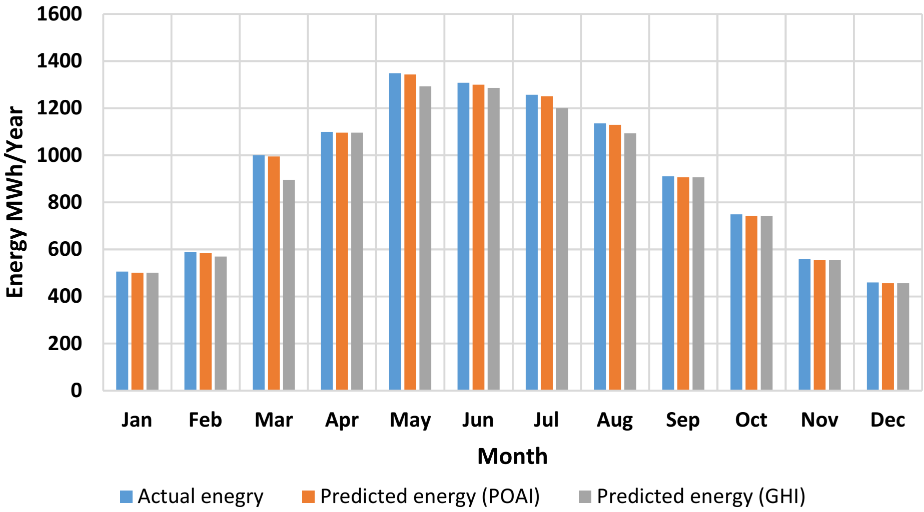


FIGURE 15 Annual trend for actual and predicted energies

TABLE 9 Geographical data of the considered sites

Location	Coordinates	Geographical region	Av. Ambient temperature	Elevation	Max. air temperature	Av. wind speed
Aswan (Benban)	27°18′0″N 33°42′0″E	Kom Imbo Desert	25.29°C	141 m	42.55	4-5 m/s
Red Sea	28°0′0″N 31°0′0″E	Hurghada	24.13°C	28 m	39.36	6-7 m/s
West and East Nile	24°30′0″N 32°42′0″E	Al Minya Desert	21.3°C	194 m	40.52	6-7 m/s

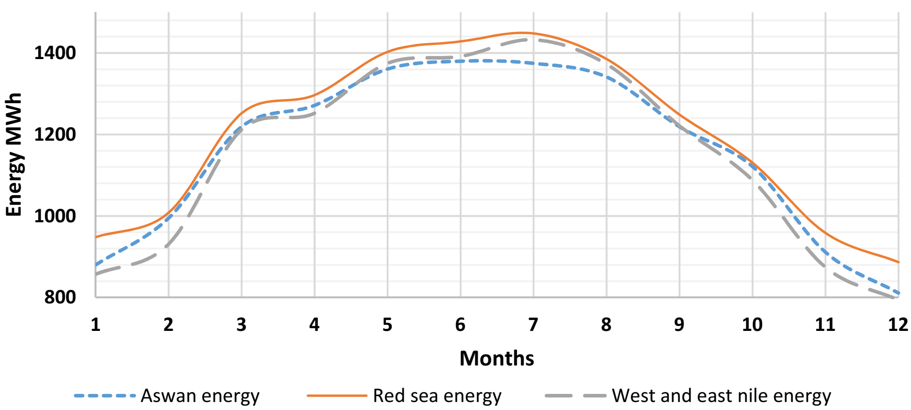


FIGURE 16 Distribution of estimated energy production for the three locations in Egypt

TABLE 10 Comparison between Egypt's three locations

Parameters	Aswan	Red Sea	West and East Nile
Plant net energy GWh/year	13.88	14.39	13.8
POA insolation kWh/m^2/year	2788.90	2913.20	2732.43
PR%	73.74	73.19	74.82
CF%	35.21	36.51	35.01

TABLE 11 Loss parameters for Egypt's locations

Location	IAM factor on global %	Module temperature %	Module mismatch %	DC wiring loss %	Inverter efficiency %
Aswan	−1.73	−9.47	−0.91	−0.98	−1.94
Red Sea	−1.61	−8.89	−0.91	−0.94	−1.94
West and East Nile	−1.8	−8.03	−0.92	−0.95	−1.99

TABLE 12 Devoted lands for Egypt's three locations

Location	Name	Acres	km^2
Red Sea	HURGHADA LOCATION 1 Al-Ahyaa	8095.238	34
Red Sea	HURGHADA LOCATION 2 KILO-10	595 238.09	2500
Aswan	BENBAN LOCATION	8843.28	536.72
Aswan	FARES LOCATION	3621.2	15.212
West and East Nile Location	LAND NO.3	109 897.11	461.567862
West and East Nile Location	LAND NO.2	338 325.49	1420.967
West and East Nile Location	LOCATION-1	127 790.54	536.72

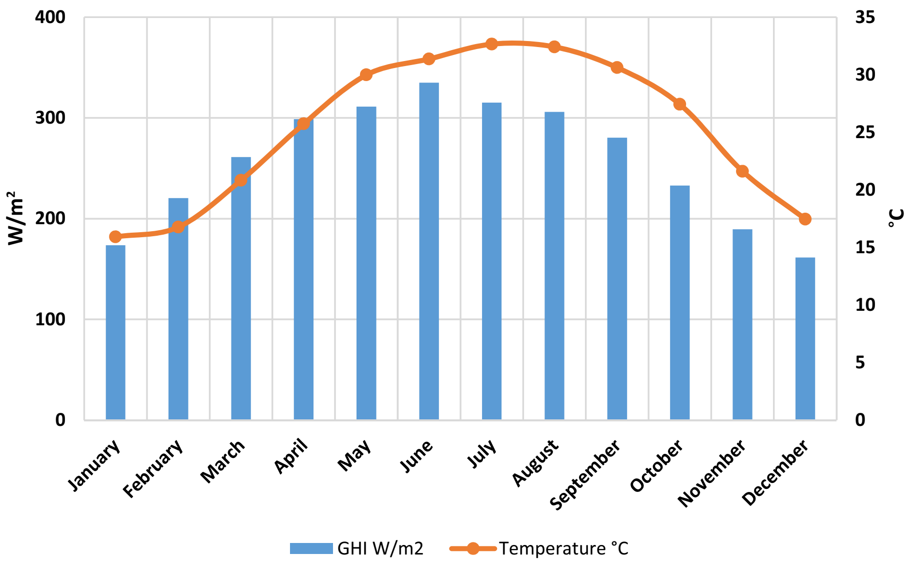


FIGURE 17 Monthly average distribution of GHI and ambient temperature for Egypt's location

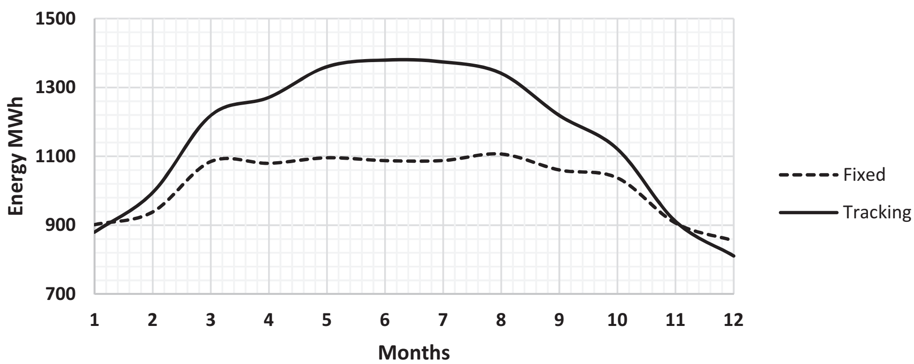


FIGURE 18 Comparison between fixed and tracking technology

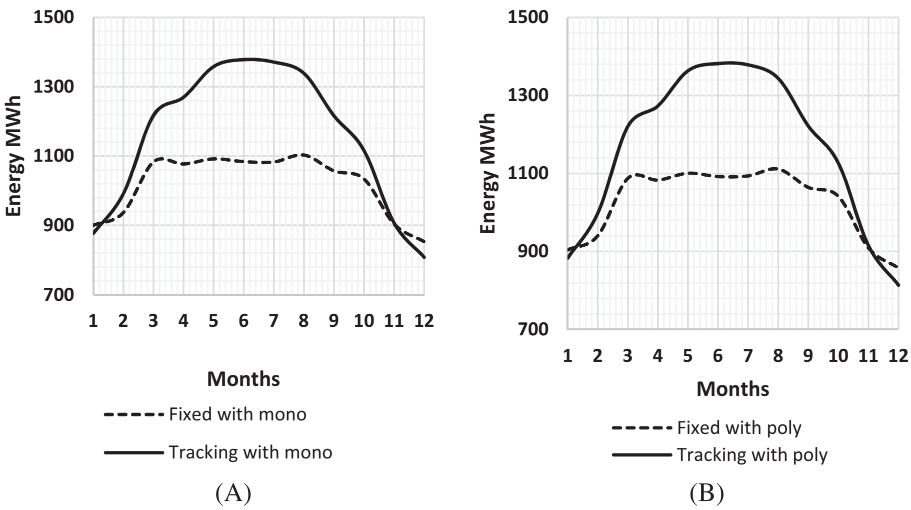


FIGURE 19 Comparing mounting technology with respect to the module type, A, Fixed and tracking technologies using monocrystalline module type, B, Fixed and tracking technologies using polycrystalline module type

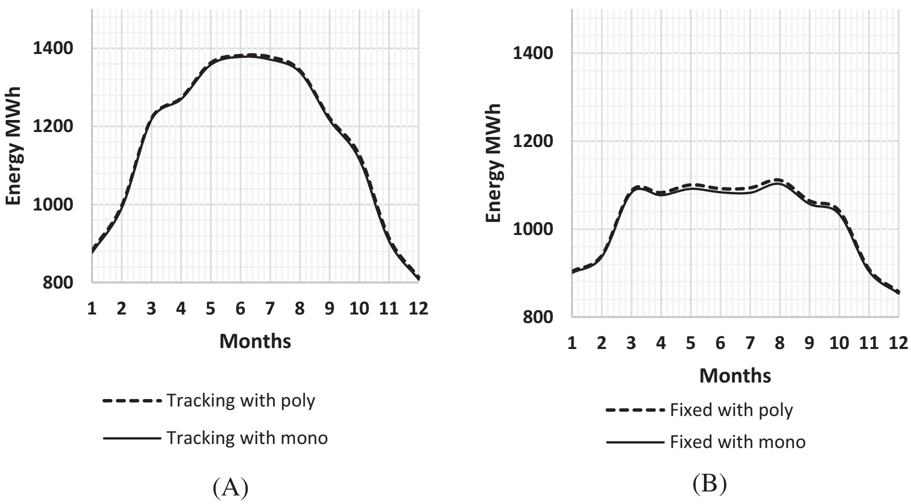


FIGURE 20 Comparing module type with respect to the mounting technology, A, using polycrystalline module and monocrystalline module (using tracking technology); B, using polycrystalline modules and monocrystalline module (using fixed mounting)

TABLE 13 Egypt's estimated output using fixed and tracking mounts

Parameters	Fixed	Tracker	% of increase
Plant net energy GWh/year	12.24	13.88	13.39
POA insolation kWh/m^2/year	2424.87	2788.90	
PR%	74.79	73.74	
CF%	31.05	35.21	

TABLE 14 Estimated outputs using fixed mounting and tracking technology (monocrystalline module)

Parameters	Fixed with mono	Tracking with mono	Percentage of increase
Plant net energy GWh/year	12.21	13.85	13.4%
POA insolation kWh/m^2/year	2424.8	2789.4	
PR%	74.57	73.56	
CF%	30.96	35.13	

TABLE 15 Estimated output using fixed mounting and tracking technology (polycystalline module)

Parameters	Fixed with poly	Tracking with poly	Percentage of increase
Plant net energy GWh/year	12.29	13.9	13.26%
POA insolation kWh/m^2/year	2424.8	2789.4	
PR%	75.06	73.93	
CF%	31.17	35.31	

TABLE 16 Estimated outputs using monocrystalline module and polycrystalline module (tracking technology)

Parameters	Tracking with mono	Tracking with poly	Percentage of increase
Plant net energy GWh/year	13.85	13.92	0.5%
POA insolation kWh/m^2/year	2789.4	2789.4	
PR%	73.56	73.93	
CF%	35.13	35.31	

TABLE 17 Estimated outputs using monocrystalline module and polycrystalline module (fixed mounting)

Parameters	Fixed with mono	Fixed with poly	Percentage of increase
Plant net energy GWh/year	12.21	12.29	0.6%
POA insolation kWh/m^2/year	2424.87	2424.87	
PR%	74.57	75.06	
CF%	30.96	35.13	

TABLE 18 Energy predicted loss factors for fixed mounting

Module technology	IAM factor on global %	Module temperature %	Module mismatch %	DC wiring loss %	Inverter efficiency %
Poly	−2.51	−8.47	−0.91	−0.92	−1.92
Mono	−2.51	−9.04	−0.91	−0.96	−1.95

TABLE 19 Energy predicted loss factors for tracking technology

Module technology	IAM factor on global %	Module temperature %	Module mismatch %	DC wiring loss %	Inverter efficiency %
Poly	−1.73	−9.16	−0.91	−0.96	−1.93
Mono	−1.73	−9.78	−0.91	−1	−1.95

4 | CONCLUSION

In this study, performance analysis for 4.5 MW$_p$ grid-connected photovoltaic farm "Gold Tree Solar Farm" located at the California Polytechnic State University, San Luis Obispo, CA, United States is conducted. This study investigated the modeling accuracy when using POAI as well as GHI irradiance measurements with the introduced approach. Cases are performed for 2019 using actual data from the farm after real-time monitoring for the site and performing performance analysis using five different parameters. Once the software simulation model is validated, the model is used to perform analyses for different locations in Egypt to estimate their energy generation potential. Finally, parameters such as tracking and module types are varied to check the best technologies for Egypt's weather.

In conclusion, the accumulated PR for the whole year of 2019 is 79% and the total energy injected into the grid is 10.92 GWh/year, with an annual capacity factor of 27.7%. The system reaches its peak in January at an accumulated PR of 92%, while PR in July is 71%. However, July has the peak energy and January has the lowest energy production. The correlation of the parameters helps us understand the effect of each parameter during the full year and shows the direct and indirect effect of the parameters on the farm system. The site has an accepted range of PR in all seasons between 75% and 95% and 79% for the whole year; it is observed that the site performs with an acceptable range of PR and even more in cold weather. The relative error ranging from 0.78% to 5.8% in all seasons and 0.54% for the whole year when using POAI, while 5% to 6.9% in all seasons and 2.9% for the whole year when using GHI. It is concluded that the new approach of combining DIRINT and Perez models to optimize the designed model gives the lowest relative error of 2.9%.

The power generation prediction of Egypt's locations shows that the Red Sea has the highest energy potential with 14.39 GWh/year. This energy potential is 13.39% per year more than Aswan and West and East Nile locations. The analysis of some parameters shows that the polycrystalline module type is more favorable and performed better for Egypt's weather conditions. It gives higher energy over the monocrystalline type with an increase in energy of 0.5% per year. It is shown for the pilot model that single-axis tracking technology gives an estimated extra energy of 1.64 GWh/year compared with the fixed mount technology as guidance for future work, depending on the cost analysis and the budget of the project. The investigators recommend the Red Sea location for the decision-makers in Egypt. It is one of the best locations for installing a new PV power plant in the future. The system should use polycrystalline modules type as well as a single-axis tracking technology.

4.1 | Future scope

For future work, an economical and environmental impact assessment for PV plants can be conducted for the proposed location to maximize profitability. Furthermore, lifetime environmental impact assessment for plants includes emission calculation. More research related to the prediction model should be investigated to increase the accuracy of the prediction that could be done by introducing more parameters in the weather file by checking the effect of the module temperature on prediction accuracy. Introducing cloud parameter usage integrated with "PlantPredict" software using Sky Camera for short-term predictions of local solar irradiance. Additionally, design and implement a pumping system in "PlantPredict" using API system. Moreover, the investigation of sub-hourly energy generation prediction for the installed PV power plants in Egypt.

NOMENCLATURE

AC	alternating current
API	application programming interface
AOI	the angle of incidence between the sun and the plane of the array
AUX	auxiliary
Cal Poly	California PolyTechnic State University
CF	capacity factor
DC	direct current
DHI	diffused horizontal irradiance
$DIRINT_{coeff.}$	DIRINT coefficient
DNI	direct normal irradiance
E_a	extraterrestrial radiation
F_1 & F_2	brightness coefficients
f_d	the diffuse fraction
G_o	extraterrestrial irradiance
GHI	global horizontal irradiance
k	is a constant equal to 1.041 for angles are in radians, or 5.535×10^{-6}
K_n	direct normal transmittance
K_{nc}	the clear-sky normal transmittance
K_T	clearness index
MAE	mean absolute error
m_a	the air mass at the sea level
$m_{a,p}$	pressure corrected air mass
Mono	monocrystalline type of module
POA	plan of array
POAI	plan of array irradiance
$POAI_{Diffused\ tilted}$	tilted diffused irradiance
$POAI_{Reflected}$	surface reflected irradiance
$POAI_{Tilted}$	tilted beam irradiance
PR	performance ratio
PV	photovoltaic
RMSE	root mean square error
R	coefficient of correlation
β	tilt angle
θ	incidence angle
θ_T	the array tilt angle from horizontal
θ_Z	the solar zenith angle
ρ_{gnd}	ground Albedo
ε	the sky clearness

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REFERENCES

1. Nathan AJ, Scobell A. How China sees America. *Foreign Aff*. 2012;91(5):1689-1699. doi:10.1017/CBO9781107415324.004
2. Nassar IA, Hossam K, Abdella MM. Economic and environmental benefits of increasing the renewable energy sources in the power system. *Energy Rep*. 2019;5:1082-1088. doi:10.1016/j.egyr.2019.08.006
3. International Renewable Energy Agency. (2018). Renewable energy outlook: Egypt. Retrieved from https://www.irena.org/publications/2018/Oct/Renewable-Energy-Outlook-Egypt
4. Kaur A, Nonnenmacher L, Pedro HTC, Coimbra CFM. Benefits of solar forecasting for energy imbalance markets. *Renew Energy*. 2016;86:819-830. doi:10.1016/j.renene.2015.09.011
5. Lolla PR, Rangu SK, Dhenuvakonda KR, Singh AR. A comprehensive review of soft computing algorithms for optimal generation scheduling. *Int J Energy Res*. 2021;45(2):1170-1189. doi:1002/er.5759
6. Rangu SK, Lolla PR, Dhenuvakonda KR, Singh AR. Recent trends in power management strategies for optimal operation of distributed energy resources in microgrids: a comprehensive review. *Int J Energy Res*. 2020;44(13):9889-9911. doi:1002/er.5649
7. Kumar RS, Raghav LP, Raju DK, Singh AR. Intelligent demand side management for optimal energy scheduling of grid connected microgrids. *Appl Energy*. 2021;285(January):116435. doi:1016/j.apenergy.2021.116435
8. Sobri S, Koohi-Kamali S, Rahim NA. Solar photovoltaic generation forecasting methods: a review. *Energy Convers Manag*. 2018;156(December 2017):459-497. doi:1016/j.enconman.2017.11.019
9. Wijeratne WMPU, Yang RJ, Too E, Wakefield R. Design and development of distributed solar PV systems: Do the current tools work? *Sustain Cities Soc*. 2019;45:553-578. doi:1016/j.scs.2018.11.035
10. Chikh M, Mahrane A, Bouachri F. PVSST 1.0 sizing and simulation tool for PV systems. *Energy Procedia*. 2011;6:75-84. doi:1016/j.egypro.2011.05.009
11. Irwan YM, Amelia AR, Irwanto M, et al. Stand-alone photovoltaic (SAPV) system assessment using PVSYST software. *Energy Procedia* Elsevier B.V. 2015;79:596-603. doi:1016/j.egypro.2015.11.539
12. Kumar NM, Kumar MR, Rejoice PR, Mathew M. Performance analysis of 100 kWp grid connected Si-poly photovoltaic system using PVsyst simulation tool. *Energy Procedia*. 2017;117:180-189. doi:1016/j.egypro.2017.05.121
13. Monsalve P, Royal KTH, Capobianchi E, Österlund B. ScienceDirect evaluation methods for photovoltaic installations on existing buildings at the KTH campus in Stockholm, Sweden assessing the feasibility of using the heat demand-outdoor temperature function for a district heat. *Energy Procedia*. 2017;115:409-422. doi:1016/j.egypro.2017.05.038
14. Malvoni M, Leggieri A, Maggiotto G, Congedo PM, De Giorgi MG. Long term performance, losses and efficiency analysis of a 960 kWP photovoltaic system in the Mediterranean climate. *Energy Convers Manag*. 2017;145:169-181. doi:1016/j.enconman.2017.04.075
15. Pierro M, De Felice M, Maggioni E, et al. Data-driven upscaling methods for regional photovoltaic power estimation and forecast using satellite and numerical weather prediction data. *Sol Energy*. 2017;158(May):1026-1038. doi:10.1016/j.solener.2017.09.068
16. Shaari S, Bowman N. Photovoltaics in buildings: a case study for rural England and Malaysia. *Renew Energy*. 1998;15(1–4):558-561. doi:10.1016/s0960-1481(98)00224-9
17. Moslehi S, Reddy TA, Katipamula S. Evaluation of data-driven models for predicting solar photovoltaics power output. *Energy*. 2018;142:1057-1065. doi:10.1016/j.energy.2017.09.042
18. Roumpakias E, Stamatelos A. Comparative performance analysis of grid-connected photovoltaic system by use of existing performance models. *Energy Convers Manag*. 2017;150(July):14-25. doi:10.1016/j.enconman.2017.08.001
19. Cubukcu M, Gumus H. Performance analysis of a grid-connected photovoltaic plant in eastern Turkey. *Sustain Energy Technol Assess*. 2020;39(March):100724. doi:10.1016/j.seta.2020.100724
20. Kumar M, Chandel SS, Kumar A. Performance analysis of a 10 MWp utility scale grid-connected canal-top photovoltaic power plant under Indian climatic conditions. *Energy*. 2020;204:117903. doi:10.1016/j.energy.2020.117903
21. Mensah LD, Yamoah JO, Adaramola MS. Performance evaluation of a utility-scale grid-tied solar photovoltaic (PV) installation in Ghana. *Energy Sustain Dev*. 2019;48:82-87. doi:10.1016/j.esd.2018.11.003
22. Monteiro GP, Palmero-Marrero AI, Moreira C, Oliveira AC. Evaluation of the performance of a photovoltaic power plant installed in a building in the north of Portugal. *Energy Procedia*. 2018;153:42-47. doi:10.1016/j.egypro.2018.10.033
23. Chawla S, Tikkiwal VA. Performance evaluation and degradation analysis of different photovoltaic technologies under arid conditions. *Int J Energy Res*. 2021;45(1):786-798. doi:10.1002/er.5901
24. Georgitsioti T, Pearsall N, Forbes I, Pillai G. A combined model for PV system lifetime energy prediction and annual energy assessment. *Solar Energy*. 2019;183(July 2018):738-744. doi:10.1016/j.solener.2019.03.055
25. de Lima LC, de Araújo Ferreira L, de Lima Morais FHB. Performance analysis of a grid connected photovoltaic system in northeastern Brazil. *Energy Sustain Dev*. 2017;37:79-85. doi:10.1016/j.esd.2017.01.004
26. Sharma V, Chandel SS. Performance analysis of a 190kWp grid interactive solar photovoltaic power plant in India. *Energy*. 2013;55:476-485. doi:10.1016/j.energy.2013.03.075
27. Kymakis E, Kalykakis S, Papazoglou TM. Performance analysis of a grid connected photovoltaic park on the Island of Crete. *Energ Conver Manage*. 2009;50(3):433-438. doi:10.1016/j.enconman.2008.12.009
28. Ayompe LM, Duffy A, McCormack SJ, Conlon M. Measured performance of a 1.72 kW rooftop grid connected photovoltaic system in Ireland. *Energy Convers Manag*. 2011;52(2):816-825. doi:10.1016/j.enconman.2010.08.007

29. Sidrach-de-Cardona M, Mora López L. Performance analysis of a grid-connected photovoltaic system. *Energy*. 1999;24(2):93-102. doi:10.1016/S0360-5442(98)00084-X
30. Ibrahim IM, Shehata AS, Shehata AI, Mohamed A. Performance analysis of 10MWp grid-connected photovoltaic system in the Mediterranean climate using PVsyst software. *Int J Smart Grid Clean Energy*. 2021;10(4):286-291. doi:10.12720/sgce.10.4.286-291
31. Erbs DG, Klein SA, Duffie JA. Estimation of the diffuse radiation fraction for hourly, daily and monthly-average global radiation. *Solar Energy*. 1982;28(4):293-302. doi:10.1016/0038-092X(82)90302-4
32. Laboratories, S. N. (2018). No Title. Retrieved from https://pvpmc.sandia.gov/modeling-steps/1-weather-design-inputs/plane-of-array-poa-irradiance/calculating-poa-irradiance/poa-sky-diffuse/perez-sky-diffuse-model/
33. Maxwell EL. A quasi-physical model for converting hourly global horizontal to direct Normal insolation. *Sol Energy Res Instit*, (SERI/TR-215-3087). 1987. Retrieved from https://www.osti.gov/biblio/5987868
34. McCormick PG, Suehrcke H. Diffuse fraction correlations. *Sol Energy*. 1991;47(4):311-312. doi:10.1016/0038-092X(91)90123-E
35. Perez R, Ineichen P, Seals R, Michalsky J, Stewart R. Modeling daylight availability and irradiance components from direct and global irradiance. *Sol Energy*. 1990;44(5):271-289. doi:10.1016/0038-092X(90)90055-H
36. Perez RR, Ineichen P, Maxwell EL, Seals RD, Zelenka A. Dynamic global-to-direct irradiance conversion models. *ASHRAE Trans*. 1992;98(pt 1):354-369.
37. Gökmen N, Hu W, Hou P, Chen Z, Sera D, Spataru S. Investigation of wind speed cooling effect on PV panels in windy locations. *Renew Energy*. 2016;90:283-290. doi:10.1016/j.renene.2016.01.017
38. Dubey S, Sarvaiya JN, Seshadri B. Temperature dependent photovoltaic (PV) efficiency and its effect on PV production in the world—a review. *Energy Procedia*. 2013;33:311-321. doi:10.1016/j.egypro.2013.05.072
39. Kaldellis JK, Kapsali M, Kavadias KA. Temperature and wind speed impact on the efficiency of PV installations. Experience obtained from outdoor measurements in Greece. *Renew Energy*. 2014;66:612-624. doi:10.1016/j.renene.2013.12.041
40. Lazaroiu GC, Longo M, Roscia M, Pagano M. Comparative analysis of fixed and sun tracking low power PV systems considering energy consumption. *Energy Convers Manag*. 2015;92:143-148. doi:10.1016/j.enconman.2014.12.046
41. Mirzaei M, Mohiabadi MZ. A comparative analysis of long-term field test of monocrystalline and polycrystalline PV power generation in semi-arid climate conditions. *Energy Sustain Dev*. 2017;38:93-101. doi:10.1016/j.esd.2017.01.002

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